ForenSight

Explainable Multi-Signal Deepfake Forensics for Images, Videos and Audio
Hackathon Idea Submission

1. The Problem

Generative AI tools have made it very easy to create fake photos, videos and voices that look and sound real. A person's face can be swapped into a video, a voice can be cloned from a few seconds of audio, and completely fake faces can be generated in seconds. Most people cannot tell the difference by looking or listening.

This is already causing real harm:

- **Scams and fraud:** fake voice calls pretending to be a family member or bank officer asking for money.
- **Identity deception :** fake videos of celebrities, leaders or company heads promoting fake schemes.
- **Misinformation:** edited images and videos spread quickly on WhatsApp and social media.
- **Identity fraud:** face-swapped videos used to cheat online KYC checks.

Existing detection tools are mostly made for researchers. They usually work on only one type of media, give only a number like "0.83 fake" without any reason, and are hard for normal users to use. There is no simple tool where a common person can upload a file and clearly understand whether it is real or not, and why.

2. Our Solution

ForenSight is a web platform where a user can upload an image, video or audio file and get a clear result:

Result	Meaning
Likely Authentic	No Significant signs of AI generation or manipulation were detected.
AI-Generated	The whole content appears to be created by an AI model.
Manipulated	The content is real but some part of it has been changed (face swap, edited region, lip-sync, cloned voice part).
Inconclusive	Evidence is weak or mixed, so the system does not force a guess.

2.1 Main Idea: Multiple Checks Instead of One Model

Instead of depending on one single AI model, TruthLens runs several independent checks on the file. Each check looks for a different kind of clue. The results are then combined to give the final answer. This makes the system harder to fool and also makes it easy to explain the result, because we know which check found what.

2.2 How It Works

Step	What happens
1. Upload	User uploads an image, video or audio file (or pastes a link).
2. Preprocessing	File type is detected. Faces are extracted from images and video frames. Audio is separated and split into small parts.
3. Analysis	The file goes through the image, video, audio and metadata checks described below.
4. Combining results	Scores from all checks are combined into one final result and confidence score.
5. Explanation	The system lists the reasons, shows heatmaps and timelines, and writes a short summary in simple words.

6. Report and history	User can view everything on the dashboard, download a PDF report, and see past checks later.

2.3 Detection Modules

Module	What it checks	Detects
Image Analysis	AI model trained on real and fake faces/images; frequency (FFT) patterns left by image generators; error level and noise analysis to find edited areas.	Synthetic faces, AI-generated images, edited or spliced images
Video Analysis	Checks faces frame by frame; looks for flickering and changes between frames; blink rate, head movement and lighting; checks if lip movement matches the speech.	Face swaps, lip-sync fakes, edited video
Audio Analysis	Converts audio into a spectrogram and checks it with an anti-spoofing model; looks for unnatural tone, missing breathing sounds and sudden cuts.	Voice cloning, AI-generated speech, spliced audio
Metadata Check	Reads file details like camera info, editing software traces and content credentials (C2PA) if available.	Edited files, files saved by AI tools

2.4 Explaining the Result

- **Heatmap on images:** the suspicious area (for example, around the jawline in a face swap) is highlighted in colour on top of the image.
- **Timeline for video and audio:** a bar shows which seconds look fake, so the user can jump straight to that part.
- **Simple reasons:** for example, "Lip movement does not match the voice between 0:12 and 0:18" or "Skin texture near the face edge is unnaturally smooth."
- **Confidence score:** shown as a percentage, with an "Inconclusive" label when the system is not sure.

2.5 Platform Features

Feature	Description
Multi-format upload	Images: JPG, PNG, WEBP. Video: MP4, MOV, AVI. Audio: WAV, MP3, M4A.
Interactive dashboard	Result card, confidence meter, heatmap viewer, video/audio timeline and detailed evidence per check.
Simple report	Downloadable PDF with the result, main reasons, evidence images and what the user should do next.
Detection history	All previous checks saved with thumbnail, result and date, so they can be reviewed again.

2.6 Technology Stack

Part	Tools
Frontend	React, Tailwind CSS, chart and waveform libraries
Backend	Python, FastAPI
AI / ML	PyTorch, OpenCV, Librosa, pretrained models from Hugging Face
Background tasks	Celery with Redis (for long video processing)
Database	MySQL

Report	ReportLab for PDF generation
Deployment	Docker

2.7 Datasets

Media	Datasets
Images	FaceForensics++, Celeb-DF, CIFAKE, GenImage
Video	DFDC (Deepfake Detection Challenge), Celeb-DF v2
Audio	ASVspoof 2019 / 2021, WaveFake, In-the-Wild audio deepfakes

4. Why This Solution Should Exist

- **Deepfakes are increasing fast.** Voice clone scams and fake videos are already causing money loss and spreading false information, and most people are not aware of how real they look.
- **People need a simple way to check.** Anyone should be able to verify media without technical knowledge.
- **A reason matters more than a score.** Showing where and why something looks fake helps users trust the result and gives investigators real evidence.
- **One tool for all media.** Images, videos and audio are handled in one place instead of using different tools.
- **Ready for new fakes.** New AI generators keep coming. Because our system uses multiple checks, new detectors can be added easily as threats change.

5. What Makes TruthLens Different

Typical detectors	TruthLens
One model, one score	Multiple independent checks combined together
Only says real or fake	Shows where and why with heatmaps, timelines and simple reasons
Works on one media type	Works on images, videos and audio
Always gives a guess	Says "Inconclusive" when evidence is weak
Made for experts	Made for normal users with an easy report

6. Limitations and Responsible Use

- Very new AI generators may not be detected at first; the system is designed so models can be updated.
- Heavily compressed files (for example, forwarded many times) may reduce accuracy.
- TruthLens supports human decisions and does not replace them.
- Uploaded files are kept private and users can delete them anytime.

7. Conclusion

ForenSight makes deepfake detection simple and understandable. A user uploads a file, sees a clear result with reasons and evidence, and can decide with confidence whether to trust it. Our goal is to help people protect themselves from scams, fake news and impersonation.